

Classwork 08//Diffusion Process and Gaussian Distribution

Student Name: _____ **Points:** 30

A Number: _____ **Due Date:** April 07, 2026

Consider a random variable

$$\varepsilon \sim \mathcal{N}(0, \mathbf{I})$$

Imagine that $\mathbf{x}_0$ is the initial data distribution (the original data distribution).
Then, in the forward diffusion process, we have

$$\mathbf{x}_{t+1} = \sqrt{1 - \beta_t} \mathbf{x}_t + \sqrt{\beta_t} \varepsilon$$

Here, the recursive relationship collapses into a single step relative to $\mathbf{x}_0$:

$$\mathbf{x}_{t+1} = \sqrt{\bar{\alpha}_{t+1}} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_{t+1}} \varepsilon$$

where we have the following relationship:

$$\alpha_i = 1 - \beta_i, \text{ and } \bar{\alpha}_{t+1} = \prod_{i=0}^t \alpha_i$$

Suppose somebody told you that initial data $\mathbf{x}_0$ has distribution $\mathcal{N}(m, s)$ where s is the variance and m is the mean.

1. Derive the mean $\mathbb{E}[\mathbf{x}_{t+1}]$ and the variance $\mathbb{V}(\mathbf{x}_{t+1})$ in the form of β_{t+1} , m and s .
2. Also provide the expression when $\beta_{t+1} = 0.1$ for all t .
3. Derive $q(\mathbf{x}_1 | \mathbf{x}_0)$.
4. In Diffusion process, in every step, we make sure that variance of the noisy data at time t doesn't explode. This is called variance preserving method. How does it guide us in choose what β_t we can use? How is it related to stochastic differential equation (SDE)?

In doing above derivation, please state what properties of expectations, and Gaussian distributions you are using.